EXPERIMENT-1

Y.R.Yaswitha

25BAI0074

1.Finding all prime numbers in a given range using **for** loop

CODE:

```
start=int(input("Enter the starting number: "))
end=int(input("Enter the ending number: "))
for num in range(start,end+1):
    if num>1:
        for i in range(2,int(num**0.5)+1):
            if num%i==0:
                break
        else:
            print(num)
```

OUTPUT:

```
====================== RESTART: /home/anaconda/Yaswitha.py ======
Enter the starting number: 3
Enter the ending number: 79
3
5
7
11
13
17
19
23
29
31
37
41
43
47
53
59
61
67
71
73
79
```

2.Reverse a number using **while** loop

CODE:

```
num=int(input("Enter a number: "))
rev=0
while num>0:
    digit=num%10
    rev=rev*10+digit
    num//=10

print("The reversed number is: ",rev)
```

OUTPUT:

```
Type "help", "copyright", "credits" or "license()" for more information.

===================== RESTART: /home/anaconda/Yaswitha.py =====================
Enter a number: 6793467
The reversed number is:  7643976
```

3.Check whether a given number is Armstrong or not using **functions**

CODE:

```python
def armstrong(n):
    power=len(str(n))
    total=sum(int(d) ** power for d in str(n))
    return total==n
num=int(input("Enter a number: "))
if armstrong(num):
    print("The given number is an Armstrong number")
else:
    print("The given number is not an Armstrong number")
```

OUTPUT:

```
====================== RESTART: /home/anaconda/Yaswitha.py ===
Enter a number: 153
The given number is an Armstrong number

====================== RESTART: /home/anaconda/Yaswitha.py ===
Enter a number: 256
The given number is not an Armstrong number
```

4.Checking grades using **if-else** loop

CODE:

```
Name=input("Enter the name of the student: ")
Marks=int(input("Enter the marks: "))
if Marks>=90:
    print("Grade A")
elif Marks>=80:
    print("Grade B")
elif Marks>=60:
    print("Grade C")
elif Marks>=40:
    print("Grade D")
else:
    print("Fail")
```

OUTPUT:

```
====================== RESTART: /home/anaconda/Yaswitha.py ===
Enter the name of the student: Yaswitha
Enter the marks: 56
Grade D

====================== RESTART: /home/anaconda/Yaswitha.py ===
Enter the name of the student: Yaswitha
Enter the marks: 90
Grade A
```

5.Count the number of even and odd numbers in a given list of numbers using **list**

CODE:

```
numbers=list(map(int, input("Enter list elements: ").split()))
even=0
odd=0
for num in numbers:
    if num%2==0:
        even+=1
    else:
        odd+=1

print("The count of Even numbers: ",even)
print("The count of Odd numbers: ",odd)
```

OUTPUT:

```
===================== RESTART: /home/anaconda/Yaswitha.py =====================
Enter list elements: 9 8 5 6 43 3 67 90 23 12 56 78 93 45 61 69 103 105 678 99
The count of Even numbers:  7
The count of Odd numbers:  13
```

6.Search for the index of an element in the **tuple**

CODE:

```
tup=tuple(map(int, input("Enter tuple elements: ").split()))
key=int(input("Enter element: "))
if key in tup:
    print("The index of the element is: ",tup.index(key))
else:
    print("Element not in the tuple")
```

OUTPUT:

```
===================== RESTART: /home/anaconda/Yaswitha.py ===========
Enter tuple elements: 6 7 454 67 89 4 3 2 5 67  34 6 7 9999999 4 65 34
Enter element: 7
The index of the element is:  1
```

7.Attain the marks of the students using name in **dictionaries**

CODE:

```
students={"Rahul": 89,
          "Anu": 98,
          "Kiran": 78}
name=input("Enter the name of the student: ")
if name in students:
    print("Marks:",students[name])
else:
    print("Student not found")
```

OUTPUT:

```
===================== RESTART: /home/anaconda/Yaswitha.py =========
Enter the name of the student: Anu
Marks: 98

===================== RESTART: /home/anaconda/Yaswitha.py =========
Enter the name of the student: Yaswini
Student not found
```

8.Count the vowels and consonants in a **string**

CODE:

```
text=input("Enter a string: ")

vowels="aeiouAEIOU"
v=c=0

for ch in text:
    if ch.isalpha():
        if ch in vowels:
            v+=1
        else:
            c+=1

print("The count of vowels: ",v)
print("The count of consonants: ",c)
```

OUTPUT:

```
===================== RESTART: /home/anaconda/Yaswitha.py =========
Enter a string: I am a student at Vellore Institute Of Technology
The count of vowels:  17
The count of consonants:  24
```

9.Print common elements in two different **sets**

CODE:

```
A=set(map(int, input("Enter first set: ").split()))
B=set(map(int, input("Enter second set: ").split()))

print("Common Elements: ", A & B)
```

OUTPUT:

```
====================== RESTART: /home/anaconda/Yaswitha.py ========
Enter first set: 6 7 8 4 5 6 34 45 3 9
Enter second set: 6 8 34 67 88 45 888
Common Elements:  {8, 34, 45, 6}
```

10.**Exception Handling**

CODE:

```python
try:
    a = int(input("Enter first number: "))
    b = int(input("Enter second number: "))

    print("Result:", a / b)

except ZeroDivisionError:
    print("Cannot divide by zero")

except ValueError:
    print("Invalid input")
```

OUTPUT:

```
== RESTART: C:/Users/yaswi/AppData/Local/Programs/Python/Python313/yaswitha.py =
Enter first number: 89
Enter second number: 90
Result: 0.9888888888888889
>>>
== RESTART: C:/Users/yaswi/AppData/Local/Programs/Python/Python313/yaswitha.py =
Enter first number: 78
Enter second number: 0
Cannot divide by zero
>>>
```